

A micro-perturbation probe of RNA structure predictors on the CPEB3 human/chimpanzee ribozyme pair: protocol and preliminary observations

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Abstract

The human and chimpanzee CPEB3 self-cleaving ribozymes differ at exactly one nucleotide (position 30: A vs G) yet display a $\sim 4\times$ difference in self-cleavage activity *in vitro*, an activity gap attributed by Skilandat *et al.* (RNA 2016) to a P1/P1.1 helix mispairing. Both variants are the only ribozyme entries in the validation and test splits of the Stanford RNA 3D Folding benchmark. We use this pair as a *micro-perturbation probe* of two transformer-based RNA structure predictors — DRfold2 (language-model front end, single-sequence input) and RhoFold+ (Evoformer, run in single-sequence mode). We report a single empirical observation: the top-5 most-divergent residues between the DRfold2 predictions for the two variants include residues 9 and 60, the published P1 helix endpoints, while none of 5 control mutations at peripheral positions reproduces this pattern. **We are explicit that this is not a biological discovery:** the Skilandat 2016 mechanism is published, and the observed predictor behavior is parsimoniously explained by correct geometric propagation through the known long-range P1.1-P1 contact rather than by “learned mechanism”. The protocol — pair a single-nucleotide functional variant with predictions from a sequence-only structure model, then compare per-residue cascade patterns against null controls — is the contribution. It is a low-cost (< 30 min on a single GPU, no MSA) sanity check that a structure predictor responds to small input changes in a non-trivial, geometry-consistent way.

1 Scope: what this is and what it is not

This is a short methods note. To avoid overclaiming we state the scope upfront:

This work is

- An empirical observation that DRfold2’s predicted structural divergence between two near-identical sequences peaks at residues already known biochemically to be the mechanistic anchor.
- A demonstration that the same observation does not arise from five arbitrary single-nucleotide controls (suggesting the cascade is not an indiscriminate model attractor).
- A reproducible, low-cost protocol others can run on any single-nucleotide variant pair.

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This work is NOT

- A biological discovery. The P1/P1.1 mispairing mechanism was established by Skilandat *et al.* (2016).
- Evidence that the language model has “learned the mechanism” in any deep sense. The same observation is parsimoniously explained by correct geometric propagation through the documented long-range contact between position 30 (in P1.1) and residues 9/60 (P1 endpoints); any structurally well-trained predictor *should* produce this cascade.
- A statistically powered claim. With $n = 5$ control mutations, our “specificity” statement is suggestive only; a properly powered test would require ≥ 20 random single-nucleotide controls and an appropriate permutation test.
- A cross-predictor benchmark. We ran RhoFold+ in single-sequence mode for fairness with DRfold2’s input regime; this is an unfavorable configuration for RhoFold+ (which is designed for MSA input). The contrast we report is therefore architectural *within this input regime*, not an evaluation of either model’s full capability.

2 Background and motivation

The structure of an RNA molecule is set by its sequence, but the *mechanism* by which a single-nucleotide change tunes a ribozyme’s activity is rarely predictable from sequence alone. The CPEB3 ribozyme — a self-cleaving HDV-like ribozyme embedded in the second intron of the human CPEB3 gene and implicated in mammalian object-location memory [5] — provides a rare, clean test pair. Its human (R1107) and chimpanzee (R1108) variants differ at exactly one position (A30 vs G30), and Skilandat *et al.* [3] showed biochemically that the human sequence cleaves $\sim 4\times$ slower because of a P1/P1.1 helix mispairing.

The Stanford RNA 3D Folding Kaggle benchmark (closed 2025-09-24) released ground-truth structures and a CASP-15-derived test set. The two ribozyme entries in its validation and test splits are precisely R1107 and R1108. That makes the benchmark, by accident, a clean single-nucleotide perturbation probe.

What has not been examined, to our knowledge, is whether the *predicted structural response* of a SOTA RNA structure model to this 1-nt difference is concentrated at the P1 region (consistent with the biochemistry) or distributed differently. We frame the question operationally as: **does the per-residue deviation between the two predictions place residues 9 or 60 in its top-5, and does this hold across arbitrary single-nucleotide controls?**

3 Materials and methods

3.1 Targets and reference

```
R1107 (human, 69 nt):
    GGGGGCCACAGCAGAAGCGUUCACGUCGCGCCCCUGUCAGCCAUUGCACUCCGGCUGCGAAUUCUGCU
R1108 (chimp, 69 nt):
    GGGGGCCACAGCAGAAGCGUUCACGUCGCGCCCCUGUCAGCCAUUGCACUCCGGCUGCGAAUUCUGCU
                                     ^_pos 30: A->G is the
                                     only difference
```

Reference structure: PDB **7QR3** (chimpanzee CPEB3 X-ray, 2.18 Å, Przytula-Mally *et al.* [4]), chain C.

3.2 Predictors

- **DRfold2** [1]: ensemble of four configuration models; the encoder is the *RNA Composite Language Model* (RCLM), a transformer trained on unlabeled RNA sequences. No multiple sequence alignment (MSA) is required.

- **RhoFold+** [2]: run with `-single_seq_pred True` to match DRfold2’s input regime. **Caveat:** RhoFold+ is designed for MSA input; single-sequence mode is its degraded fallback, not its intended use.

3.3 Null controls

Five single-nucleotide variants of R1108 at peripheral positions:

ID	Mutation
null_pos5	pos 5 G→A
null_pos20	pos 20 U→A
null_pos41	pos 41 A→C
null_pos55	pos 55 U→A
null_pos64	pos 64 U→A

We note that these positions were selected manually as “peripheral” based on the published secondary structure; a statistically rigorous study would draw control mutations uniformly at random.

3.4 Comparison metric

For each pair (variant prediction, R1108 prediction) we Kabsch-align on C1’ atoms and compute per-residue deviation. We extract the top-5 most-divergent residues. A target “hits the P1 anchor” if residue 9 *or* residue 60 appears in its top-5. The choice of “top-5” is heuristic; results are robust to top-3 and top-7 by inspection but we did not formalize this sensitivity.

4 Results

4.1 The empirical observation

The DRfold2 predictions for R1107 and R1108 differ with a Kabsch C1’ RMSD of 1.86 Å. The per-residue deviation peaks at residues forming the P1 helix endpoints — position 9 (4.21 Å) and position 60 (4.12 Å) — and at three other central-core residues (22, 24, 51). The mutation site itself (position 30) shows a deviation of 2.15 Å, smaller than residues 9 and 60 (Table 1, row R1107_human).

4.2 The control: no NULL mutation reproduces this pattern

We applied identical inference to the five control mutations (Table 1). In none of the five does residue 9 or 60 appear in the top-5. Each control’s top-5 instead cluster *around the mutated position itself*. This is what we would naively expect from a well-behaved structure predictor: a local perturbation begets a local response.

Table 1: Multi-null comparison (DRfold2 predictions, Kabsch-aligned C1’ deviations vs the R1108 prediction). “P1?” indicates whether residue 9 or 60 is in the top-5.

Target	Pos.	mean Δ	max Δ	$\Delta@9$	$\Delta@60$	Top-5	P1?
R1107 (real)	30	1.64	4.21	4.21	4.12	{9,22,24,51,60}	YES
null_pos5	5	2.22	5.79	1.57	4.14	{1,2,3,22,23}	no
null_pos20	20	1.30	3.85	0.78	1.90	{23,24,26,48,50}	no
null_pos41	41	1.69	5.47	1.48	2.23	{48,49,50,51,52}	no
null_pos55	55	4.02	10.39	2.03	3.72	{22,23,24,25,51}	no
null_pos64	64	2.40	8.95	1.85	1.76	{50,64,65,66,67}	no

A key methodological caveat: at position 30, residue 9 shows a deviation of 4.14 Å in the `null_pos5` control as well (i.e., a P1 perturbation *does* appear when one mutates the 5' end, which itself sits in the P1 stem). Residue 9 is therefore not as “mutation-30-specific” as the top-5 framing suggests; the cleaner specificity statement is about the *simultaneous* presence of residue 9 *and* residue 60 in the top-5, which holds for the real case alone (Table 1).

4.3 Geometric distance check (Experiment #1) reduces the claim

To distinguish “the model captured mechanism” from “the model correctly modeled 3D geometry,” we measured the C1' Euclidean distance from the mutation site (residue 30) to each top-5 cascade residue in the predicted structures (Table 2). Combined with a frequency check (how often each top-5 residue appears across the five NULL control top-5s, Table 3), the picture becomes clear:

Table 2: 3D distance (Å) from C1'(pos 30) to cascade residues, in the two DRfold2 predictions for R1107 and R1108.

Predicted structure	d→9	d→22	d→24	d→51	d→60
R1107 (DRfold2)	13.37	26.75	21.16	41.28	13.84
R1108 (DRfold2)	8.02	29.73	25.96	43.33	15.71

Table 3: Frequency of each REAL top-5 cascade residue across the five NULL control top-5 lists.

Residue	In REAL top-5?	In how many NULL top-5s?	Mutation-specific?
9	yes	0 / 5	yes
22	yes	2 / 5	no (attractor)
24	yes	2 / 5	no (attractor)
51	yes	2 / 5	no (attractor)
60	yes	0 / 5	yes

Refined finding. Of the original top-5 cascade residues:

- **Residues 9 and 60** are simultaneously (a) within 15 Å of position 30 in both DRfold2 predictions (direct geometric coupling consistent with the published $P1.1 \leftrightarrow P1$ contact) AND (b) specific to the real mutation (0/5 NULL controls have them in their top-5). These are the cleanly defensible cascade residues.
- **Residues 22, 24, 51** are at long distance from position 30 (>20 Å, with residue 51 at ~41 Å) AND each appears in 2/5 NULL control top-5 lists. These are DRfold2 attractor positions, not mutation-specific responses.

The conservative reading is therefore: *DRfold2 correctly places the published $P1.1 \leftrightarrow P1$ 3D contact in its predicted structure and propagates a single-nt perturbation across that contact in a mutation-specific way.* This is a clean model-behavior observation — a sanity check on the model’s 3D coupling fidelity — not evidence that the model has internalized the mechanism beyond what geometry forces.

4.4 Secondary-structure-level evidence

ViennaRNA’s MFE folding gives different dot-bracket secondary structures for R1107 and R1108 with 40 residues differing in base-pairing state, and the human variant’s MFE is +1.00 kcal/mol higher (less stable) than the chimpanzee’s. The direction of this stability gap is consistent with the biochemical expectation that the human variant supports a weaker P1.1 helix. This is a sanity check on the secondary-structure side, not a discovery.

4.5 RhoFold+ single-sequence behaviour

In single-sequence mode (its degraded input regime), RhoFold+ reaches $\text{RMSD} = 7.7 \text{ \AA}$ vs the chimpanzee crystal. Its per-residue deviation between R1107 and R1108 is dominated by the same five residues ($\{21, 48, 49, 50, 51\}$) regardless of which mutation we apply; the top-5 for the real (pos 30) case and for the `null_pos41` control are identical. The most parsimonious interpretation is that the no-MSA architecture defaults to a fixed structural prior; this is consistent with RhoFold+ being designed to take MSAs as input. **This is not a fair comparison** of the two models’ biological insight; it is a within-input-regime architectural contrast.

5 Discussion

5.1 The parsimonious geometric explanation

The strongest competing explanation for our observation is the following. The published Skilandat 2016 model has position 30 sitting in the P1.1 segment, which is *geometrically* in contact with the P1 helix endpoints (residues 9 and 60). A correctly trained structure predictor that learns base-pair contacts and 3D proximity *should* propagate a perturbation at any position to its 3D neighbours. Therefore the cascade we observe could simply be “DRfold2 correctly models the geometry of the P1.1 $\leftrightarrow$ P1 contact,” which is a baseline expectation for any good model, not a “mechanistic insight”.

This caveat does *not* invalidate the observation. It does mean the most defensible reading is:

The empirical cascade pattern is consistent with what one would expect from a structurally well-calibrated predictor on this pair. It is also consistent with a deeper claim that the model has captured mechanism, but our experimental design cannot distinguish the two explanations.

5.2 Relevance to NLP transformer interpretability

Methodologically, micro-perturbation probing is a familiar *behavioral probing* technique in NLP: change a single token, observe whether the model’s distribution shifts in a linguistically-coherent way. Our protocol is the structure-prediction analogue. The pedagogical value is concrete: the input is unambiguous (a 1-nt swap), the prediction is high-dimensional (3D coordinates), and the expected response can be evaluated against an independent ground truth (the biochemical mispairing model). We make no claim of methodological novelty for the probing approach itself; we report a single case study.

5.3 What would constitute a real discovery

- Predicting the activity *ratio* (not just the structural delta) from sequence alone, and showing it across many variant pairs.
- Identifying a mutation/family where a predictor places a cascade at residues *not yet* described biochemically, then validating those residues experimentally.
- Showing the cascade pattern correlates with predicted confidence (pLDDT-like score) in a way that reveals where models are reliable.
- A statistically powered control study ($n=20$ random single-nucleotide variants, permutation test, multiple predictors run in their intended input regime including MSA).

We did none of the above. The protocol is offered as scaffolding for future work, not as evidence of any of these claims.

5.4 Limitations

- $n = 5$ control mutations is statistically weak.

- Only one model (DRfold2) tested as the “responsive” system.
- RhoFold+ tested only in degraded single-sequence regime.
- “Top-5” threshold is heuristic.
- No statistical permutation test on the cascade specificity.
- No use of DRfold2’s per-residue confidence channel.
- No independent validation that residues 9 and 60 are 3D-proximal to position 30 in the predicted structures (which would strengthen the geometric-propagation explanation).

6 Conclusion

We report a small, honest empirical observation: a recent single-sequence RNA structure predictor responds to the biologically functional CPEB3 single-nucleotide variant by placing its largest predicted structural changes at the published P1/P1.1 mechanistic anchor, while five peripheral control mutations elicit local responses only. We are explicit that this is consistent with — but does not require — the predictor having “learned mechanism”; a parsimonious geometric explanation suffices. The contribution we stand behind is the protocol, not a biological claim. The work is presented as scaffolding for a properly powered cross-predictor study.

7 Data and code availability

All sequences, scripts, intermediate predictions, and analysis tables are at <https://github.com/Vince-Anduril/rna3d-calibration>.

Author contributions

VLD conceived the project, made the focus-family pick (CPEB3), ran the pod sessions, validated each step, and edited the manuscript. Claude (Opus 4.7) authored the pipeline code, ran the analysis scripts, drafted the manuscript and slides under VLD’s direction, and flagged its own overclaims so they could be corrected before submission. Both authors take responsibility for the published text.

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